Use of Repo Agent and OPS Agent Opencode+Hermes

Class eight



TOOL COMPARISON AND TRADING USE CASE

Cursor vs OpenCode vs Hermes Agent

Three tools compared by role.
Then: a Hermes and OpenCode trading workflow with human approval.

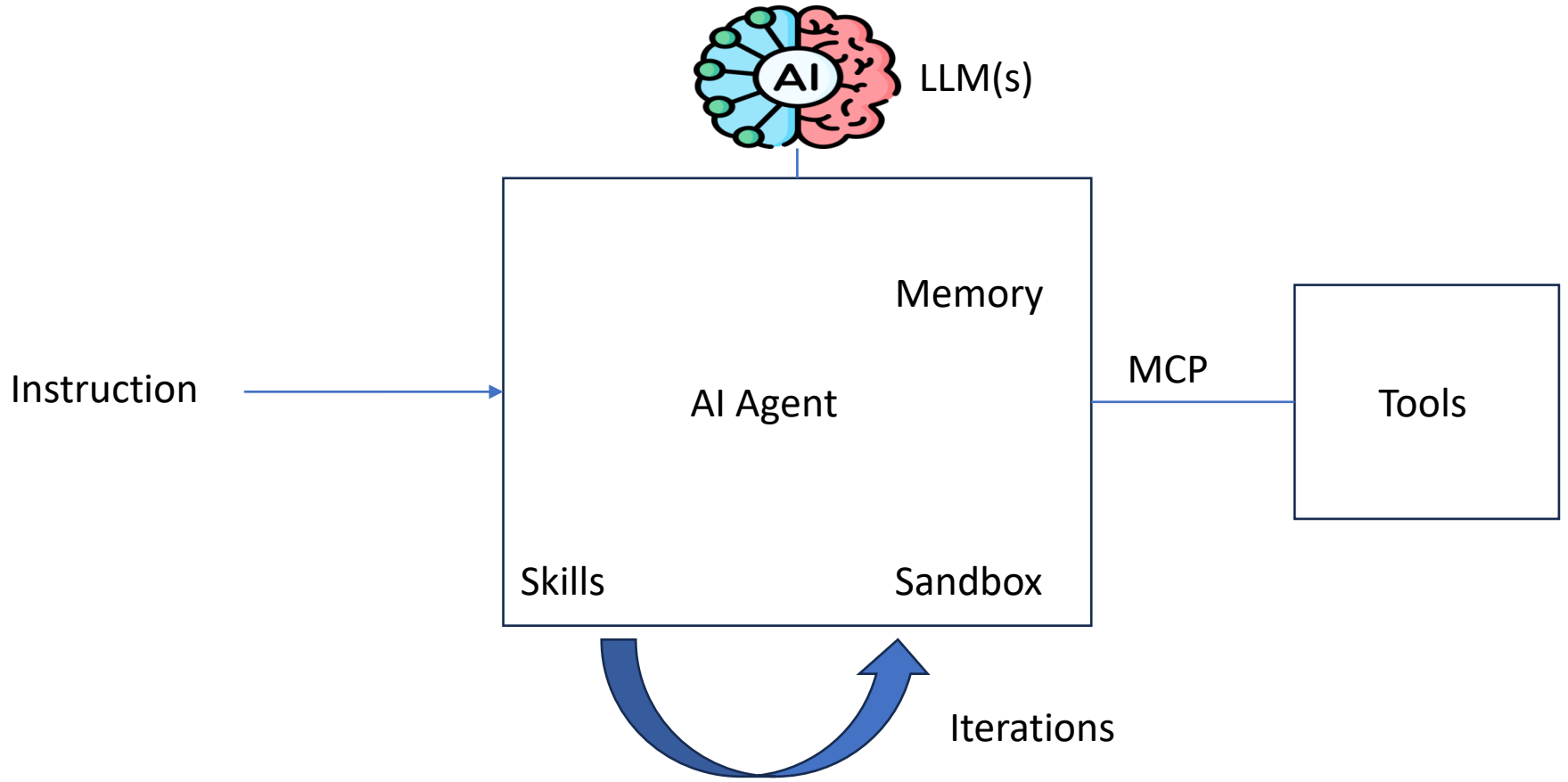
AI IDE | CODING AGENT | AGENT FRAMEWORK | IB EQ DESK

- **Cursor** — Where you sit and write
- **OpenCode** — A hired coder you send tasks to
- **Hermes** — The assistant that manages the rest — and may propose IB tickets

Agenda

- Overview of AI agents
- AI IDE, repository agent and operations agent
- OpenCode, skills and MCP
- Hermes and OpenCode trading use case
- Screening, backtesting and approved execution
- Trading diary and controlled improvement

Agentic Development Framework



AI Agent=LLM Model + Harness

Vibe Coding: Beyond Natural Language
From Fast Prototyping to Full Autonomous Development

Requirement → Vibe Prompt → AI Development → Auto Testing → Deployment → Monitoring + Docs

Classic Vibe Coding



- Natural language prompts
- AI generates/edits code
- Rapid iteration & creativity
- Human does most of the steering

Agentic Vibe Coding (The Full Power)



- Requirements → Code Generation
- Automated Testing & Debugging
- Deployment & Documentation
- Monitoring & Continuous Improvement

Vibe Coding is not just “talk to AI to write code". It's the mindset that lets you automate the entire software lifecycle using AI agents.

Common tools on AI Coding

- Cursor
- VSC+Github Copilot+LLM providers
- Opencode
- Claude code
- Hermes agent/Openclaw
- Etc

THE MAP

Same neighborhood. Different jobs.

All three use models to write and change code. The difference is the surface, the human role, and how far the system is allowed to run.

AI CODE EDITOR

Cursor

A full IDE built on VS Code. You live inside it.

SURFACE

Editor

HUMAN ROLE

In the loop — keystroke by keystroke

closed · subscription

CODING AGENT

OpenCode

A CLI / TUI agent you invoke from the terminal.

SURFACE

Terminal

HUMAN ROLE

Delegate the task, review the result

open source · MIT

AGENT FRAMEWORK

Hermes Agent

A general tool-using agent. Coding is one skill among many.

SURFACE

Terminal / desktop / chat

HUMAN ROLE

Orchestrate agents, including the others

open source · MIT

The interactive surface.

WHAT IT IS

A full IDE built on VS Code by Anysphere. You write, edit, and read code with AI embedded in the same place you already work.

HOW AI SHOWS UP

Woven into the editor: tab-complete, inline edits, chat, and agent mode. You see diffs and accept or reject them in place.

MODELS

Mostly tied to a Cursor subscription and Anthropic models. Custom API keys are possible, but the product is not model-agnostic by default.

HUMAN ROLE

In the loop at a keystroke level. The person drives. The model assists. Control stays granular.

OPEN / CLOSED

Closed-source, commercial, subscription. The editor is the product.



WHERE YOU SIT AND WRITE

Best when the work is exploratory, the repo is already open, and you want to steer every change.

Anysphere · VS Code fork

The hired coder in the terminal.

WHAT IT IS

A provider-agnostic, open-source CLI / TUI agent (opencode.ai). You invoke it from a terminal — for example, opencode run "fix this bug".

HOW IT WORKS

You hand it a task. It plans, edits files, and runs tool calls, then reports back. Long-running sessions, parallel worktrees, PR review.

MODELS

Any provider: OpenRouter, Anthropic, OpenAI, local models. You pick and swap freely. The agent is not tied to one lab.

HUMAN ROLE

You delegate and review. You are not driving at a keystroke level. There is no editor — the terminal is the surface.

OPEN / CLOSED

Open source (MIT). Community and foundation-driven rather than a closed subscription product.



A TASK YOU SEND.
A RESULT YOU REVIEW.

Best when the work is scoped, you want autonomy, and you are happy to live in a terminal instead of an IDE.

opencode.ai · MIT

The orchestrator, not just a coder.

WHAT IT IS

An open-source agent framework by Nous Research. Coding is only one of many things it does — a general tool-using agent with memory and skills.

HOW IT WORKS

Runs on terminal, a desktop app, and messaging platforms (Telegram, Discord, Slack). Spawns sub-agents. Can drive OpenCode, Codex, or Claude Code as children.

MODELS

Any provider, swappable mid-task. Credential pools rotate across multiple API keys automatically.

HUMAN ROLE

You delegate and orchestrate. Hermes manages the hired coders rather than being one of them.

OPEN / CLOSED

Open source (MIT) from Nous Research. Persistent memory and self-improving “skills.”



A GENERAL ASSISTANT THAT CAN MANAGE THE HIRED CODERS

Best when the work spans tools, channels, and memory — not just a single repo in a single session.

Nous Research · MIT

SIDE BY SIDE

At a glance

	CURSOR	OPENCODE	HERMES
Category	AI IDE	Coding agent	Agent framework
Surface	Editor	Terminal	Terminal / desktop / chat
Human role	In the loop, granular control	Delegates tasks, reviews	Delegates, orchestrates agents
Models	Subscription / Anthropic-first	Any provider, freely swapped	Any provider, mid-task swap
License	Closed, commercial	Open source (MIT)	Open source (MIT)
Maker	Anysphere	Community / foundation	Nous Research

How to keep them straight.



CURSOR

Where you sit and write.

The desk. The keyboard. The diffs you accept one at a time.



OPENCODE

A hired coder you send tasks to.

You specify the job. It plans, edits, runs, and comes back.



HERMES

A general assistant that can also manage the hired coders.

Memory, skills, sub-agents, chat surfaces — coding included.

Pick the surface, not the brand.

Reach for Cursor when

- You are already inside the repo
- You want to see every diff
- The work is exploratory or tightly coupled to the editor
- Granular accept / reject matters

Reach for OpenCode when

- The task is scoped enough to hand off
- You want provider freedom or local models
- You prefer terminal over IDE chrome
- You need long sessions or parallel worktrees

Reach for Hermes when

- The job is bigger than one repo
- You need memory that survives a session
- Work should land in Slack, Discord, or Telegram
- Other coding agents should be children, not peers

Opencode

Installation of Opencode

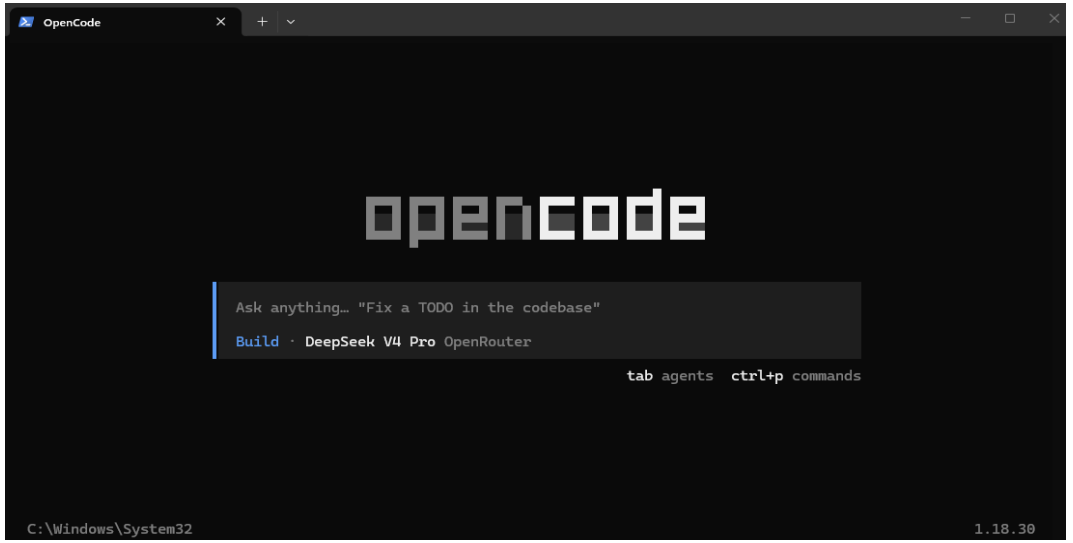
- https://opencode.ai/



curl -fsSL https://opencode.ai/install | bash (Mac)

npm i -g opencode-ai

You need to install node.js and npm



Two execution modes

Run mode

```
PS D:\Algo_development_2026\web scraping> opencode run --auto "how you see the price of gold"
```

```
Unexpected token ')' in expression or statement.
At line:1 char:167
+ ... .get('regularMarketPrice', info.get('previousClose', 'N/A'))}\"); pri ...
+                                                                  ~
Unexpected token '}' in expression or statement.
    + CategoryInfo          : ParserError: (:) [], ParentContainsErrorRecordException
    + FullyQualifiedErrorId : MissingExpressionAfterToken

← Write C:/Users/user/AppData/Local/Temp/opencode/gold_price.py
Wrote file successfully.

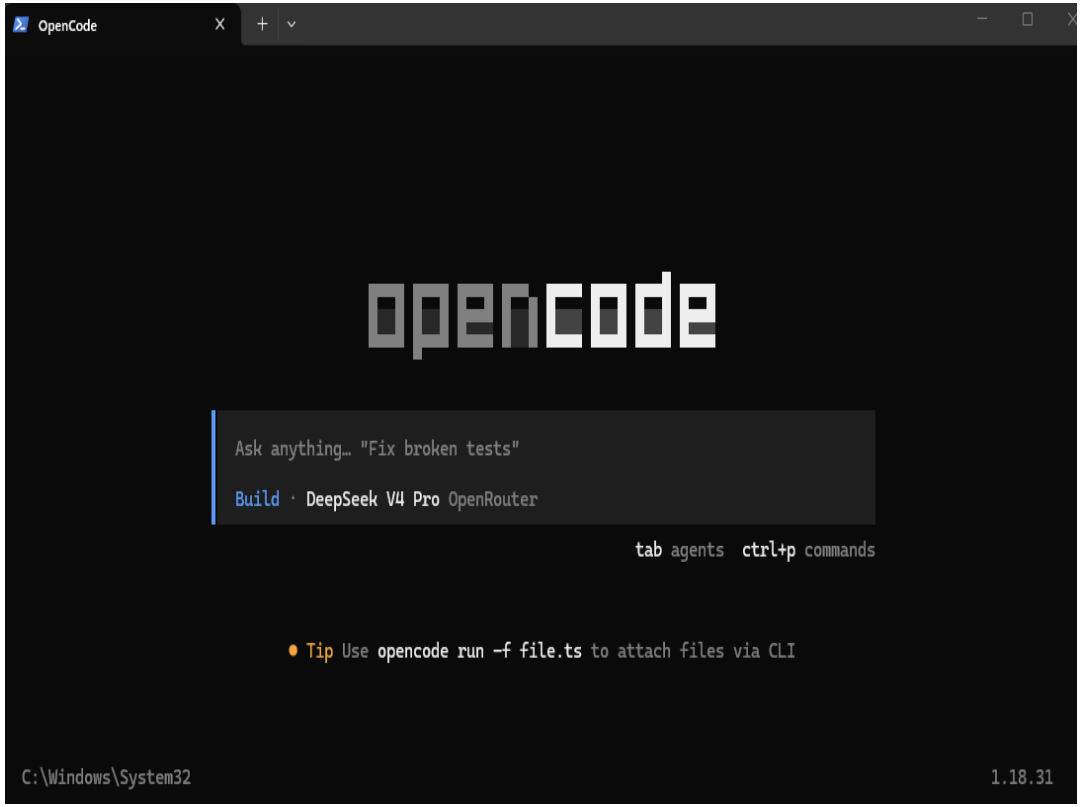
$ python "C:\Users\user\AppData\Local\Temp\opencode\gold_price.py"
Name: Gold Dec 26
Current Price: $4364.5
Bid: $4365.4
Ask: $4365.7
Day Range: $4294.5 - $4379.7
50-Day Avg: $4320.614

5-Day History:
  2026-09-14  Close: $4351.90
  2026-09-15  Close: $4332.80
  2026-09-16  Close: $4387.50
  2026-09-17  Close: $4364.50

Gold Dec 26 futures (GC=F) are currently at **$4,364.50** per ounce, with a day range of $4,294.50 - $4,379.70. Over the
 past 5 days, gold has traded between ~$4,333 and $4,388.

PS D:\Algo_development_2026\web scraping>
```

TUI mode



Setup LLM models

- /connect
- Select the vendors of the LLM providers (e.g. Openrouter)
- Select the model (e.g. Deepseek-V4-Pro)

Use Case

Telegram bot

Telegram bridge (Python)

http://127.0.0.1:4096

Repo (Project folder)

Opencode headless server

Agent.md
Skills

LLM
Provider

Include python scripts:
-fetching stock data
-telegram bridge

Python scripts

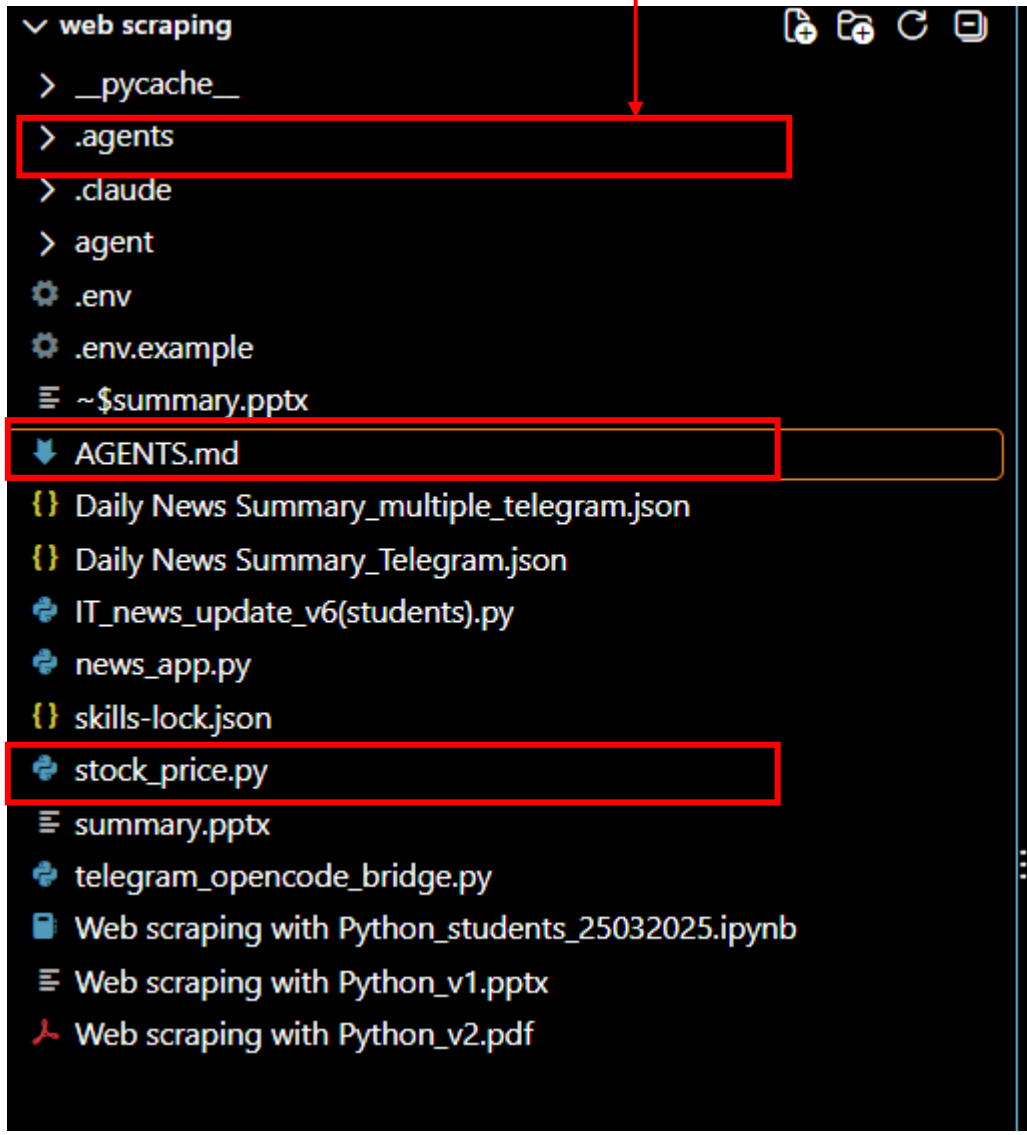


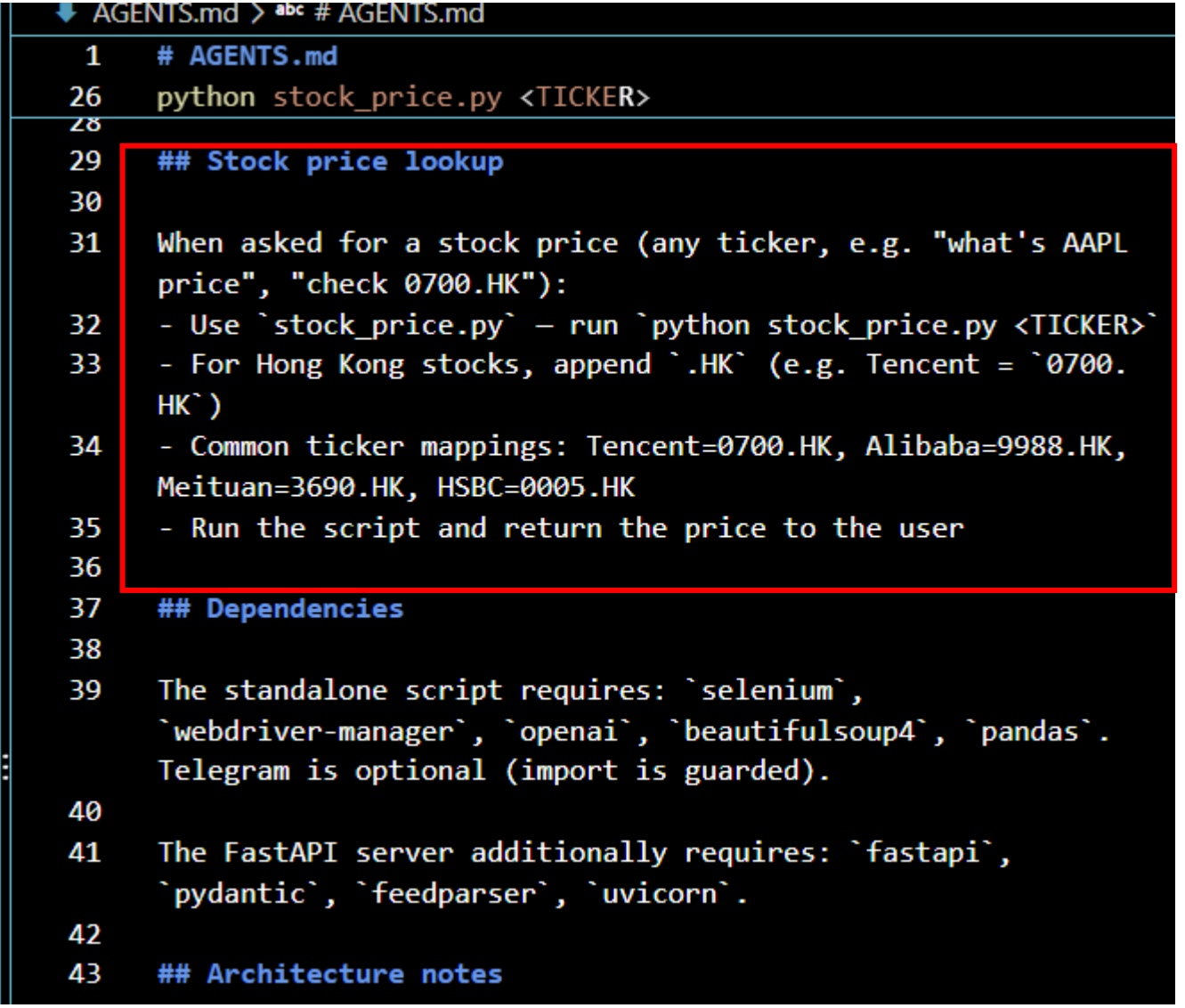
Under the project folder

```
PS D:\Algo_development_2026\web scraping> opencode serve --port 4096 --hostname 127.0.0.1
Warning: OPENCODE_SERVER_PASSWORD is not set; server is unsecured.
opencode server listening on http://127.0.0.1:4096
```

Agents.md

Skills





Skills

Skills tell OpenCode how to work



Example:Install Skills under Opencode

- npx skills add HKUDS/Vibe-Trading --skill vibe-trading -a opencode -y

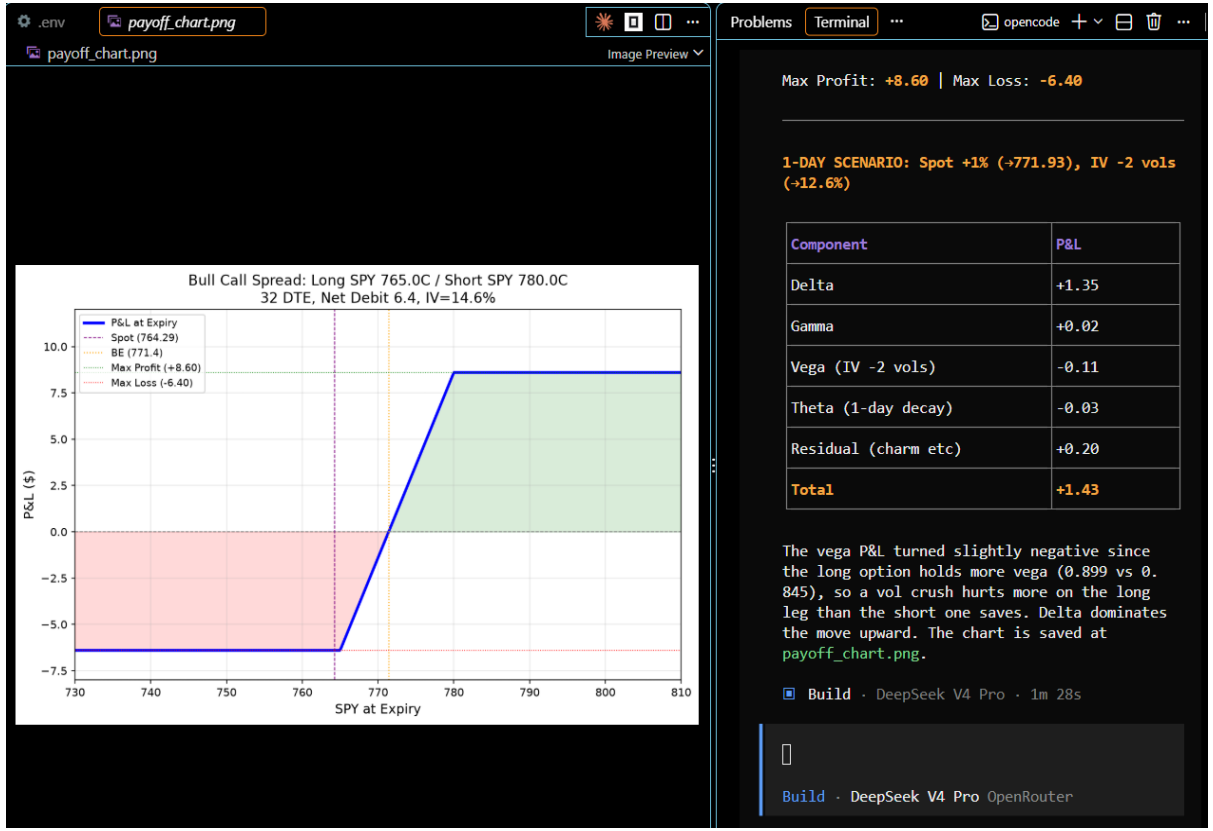
Example: Using the Skills under opencode

Load options-payoff.

Structure: long SPY 765 call / short SPY 780 call, net debit 6.40, expiry 32 DTE, spot 764.29.

Draw payoff at expiry and a table of P&L at spot 740, 755, 765, 771.40, 780, 800.

Add a 1-day scenario: spot +1%, IV −2 vols. Show delta/gamma/theta/vega P&L split.

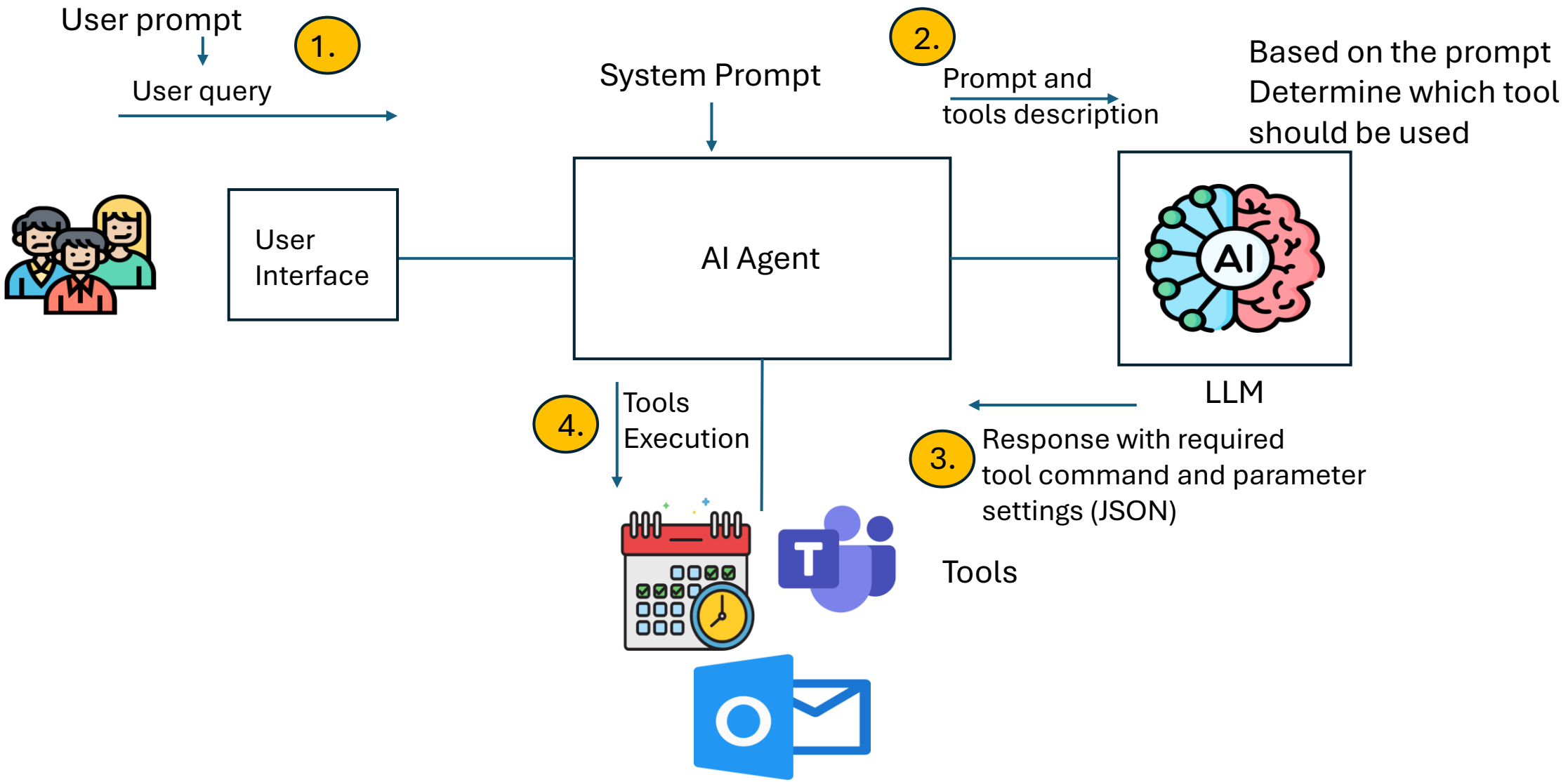


MCP

MCP tools give it live connectors



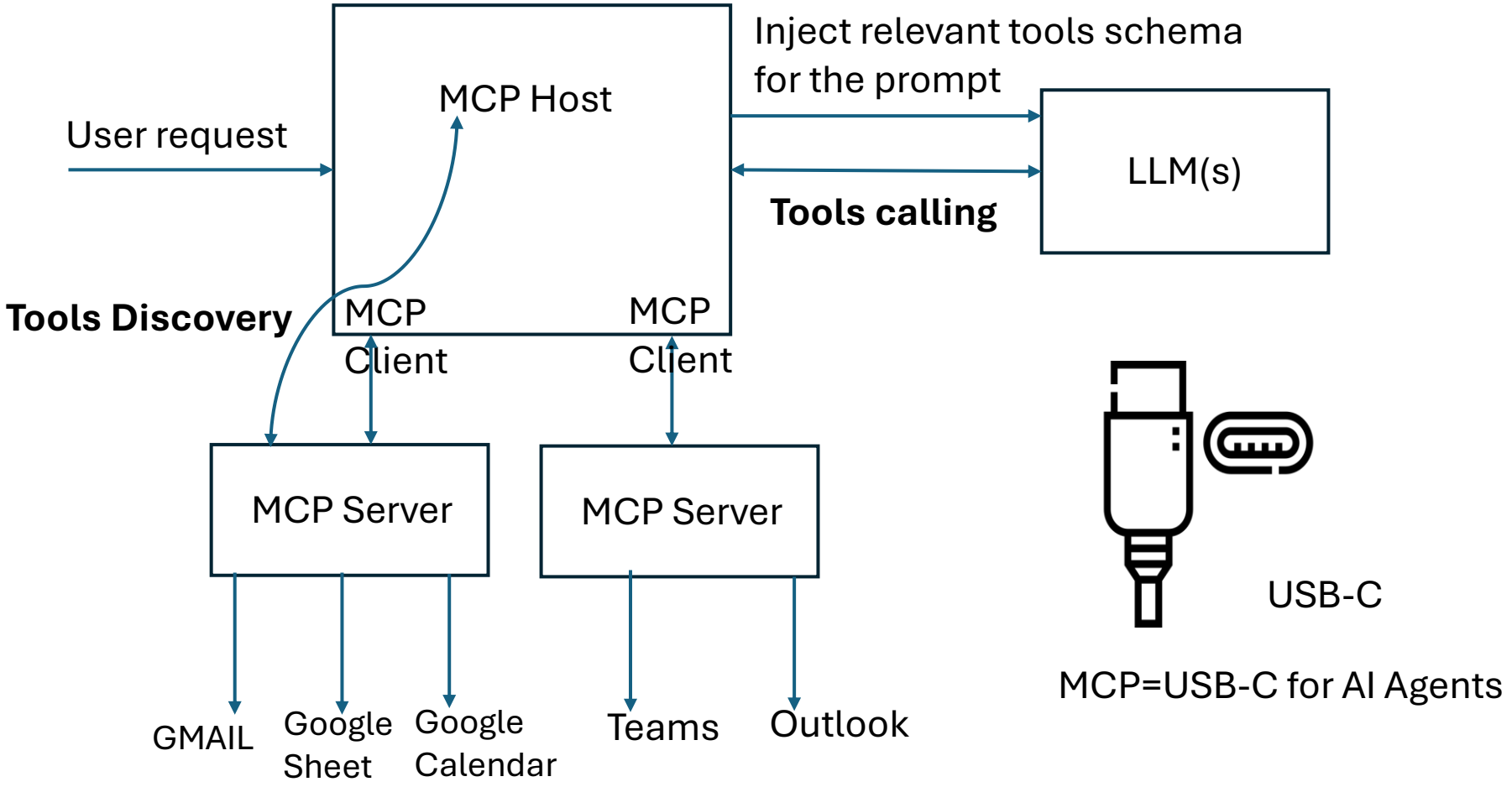
Tools Calling



Components in MCP

Component	Role	Examples
MCP Host	The user-facing AI application (e.g., Claude Desktop, an IDE like Zed or Cursor, Copilot Studio, a custom agent app)	Chat interface, coding assistant
MCP Client	Embedded in the host; speaks the MCP protocol to servers	Handles requests from the LLM
MCP Server	Lightweight service that exposes tools, resources, or prompts	GitHub MCP server, SQL database server, local file system server

MCP (Model Context Protocol)
Standardize the way AI applications work with LLM



Benefits of MCP

MCP is the “USB-C for AI agents”:

- Instead of building a custom cable for every new data source or LLM, you plug in standardized MCP servers, and everything just works — securely, privately, and across the entire AI ecosystem.
- One protocol to rule them all:
  - Instead of writing separate code for GitHub, Slack, Google Drive, a database, etc., you implement MCP once on the server side.
  - AI agents can dynamically discover, inspect, and use tools/resources without custom glue code.

Benefits of MCP

Benefit	Explanation	When It Matters Most
Modularity & Reusability	Tools live on separate servers. You can update, improve, or fix a tool without touching your agent.	Teams, production systems, long-term projects
Dynamic Tool Plugging	**Add new tools/servers on the fly without redeploying your workflow.**	Complex agents that need evolving capabilities
Standardization	One standard works across Claude, OpenAI, Grok, local models, n8n, etc.	Multi-model setups
Better Ecosystem	**Huge and growing library of ready-made MCP servers** (Git, Files, Browser, Databases, APIs, etc.)	Rapid development
Security & Control	**Tools run on their own servers with proper auth, permissions, and auditing.**	Enterprise / sensitive data
Discoverability	**Tools can describe themselves (capabilities, resources, prompts).**	Advanced agents
Separation of Concerns	**AI logic stays separate from tool logic. Different teams can own different parts.**	Large projects

Install MCP tools
Example: Install Playwright MCP tools

1. Install @anthropic/mcp-server-playwright:

npm install -g @anthropic/mcp-server-playwright2.

2. Add to your opencode.json (in your project root or ~/.config/opencode/):

{ "$schema": "https://opencode.ai/config.json", "mcp": { "playwright": { "type": "local", "command": ["npx", "-y", "@anthropic/mcp-server-playwright"], "enabled": true } }}

3. Restart opencode

Example: Use MCP

Playwright MCP: open

https://inews.hket.com/sran009/%E5%8D%B3%E5%B8%82%E8%B2%A1%E7%B6%93?mtc=80042

Snapshot the 即市財經 list, dismiss popups, scroll twice.

Extract up to 15 unique /article/ items (dedupe by id).

For each: verbatim 標題, 時間, 2-句摘要 from the teaser, full URL.

Traditional Chinese. Table only. Do not invent headlines.

1	【美股開市】AI股受壓、英偉達挫4% 油價雙雙破百（不斷更新）	今日 21:44	油價急升雙雙破百，紐約期油升3.15%報103.2美元，布蘭特期油升3.52%報108.29美元。 沙特東西輸油管遭攻擊損壞，未來數周大部分時間停運。	https://inews.hket.com/article/4193183 (https://inews.hket.com/article/4193183)
2	【伊朗局勢】紐油再升4%、布油逼110美元關 胡塞武裝再奪紅海戰略島嶼　路透：沙特關鍵輸油管停運、恐削全球供應4%	今日 21:35	也門胡塞武裝進一步佔領紅海南部多個戰略島嶼，控制範圍距美軍基地僅約32公里。沙特東西輸油管道遭無人機襲擊關閉，維修需時三至五周。	https://inews.hket.com/article/4191750 (https://inews.hket.com/article/4191750)
3	【Kioxia】據報鎧俠擬美國發ADR籌100億美元 已與美銀高盛摩通洽談	今日 21:11	日本記憶體公司鎧俠擬透過發ADR籌100億美元，開始與美銀、高盛、摩通等投行討論明年IPO。若成功發ADR，有可能被納入費城半導體指數。	https://inews.hket.com/article/4193468 (https://inews.hket.com/article/4193468)
4	【王健林旗下】HYROX易手！　盈方體育據報以54億出售HYROX 多數股權　LVMH旗下私募基金接手	今日 19:27	萬達旗下盈方體育向LVMH旗下私募基金L Catterton牽頭財團出售HYROX 多數股權，作價約6億歐元。HYROX 為結合跑步與功能性訓練的室內體能耐力賽事。	https://inews.hket.com/article/4193443 (https://inews.hket.com/article/4193443)

Hermes and OpenCode trading architecture

HERMES

Workflow coordinator

Schedules approved jobs, summarizes results and requests your approval through Telegram. Uses restricted tools.

OPENCODE

Development agent

Implements Python screens, backtests and broker adapters in a development branch. Runs tests and submits changes for review.

PYTHON SERVICES

Calculation and execution

Validated code computes signals and risk, checks approval, submits orders and records broker events. No live code rewriting.

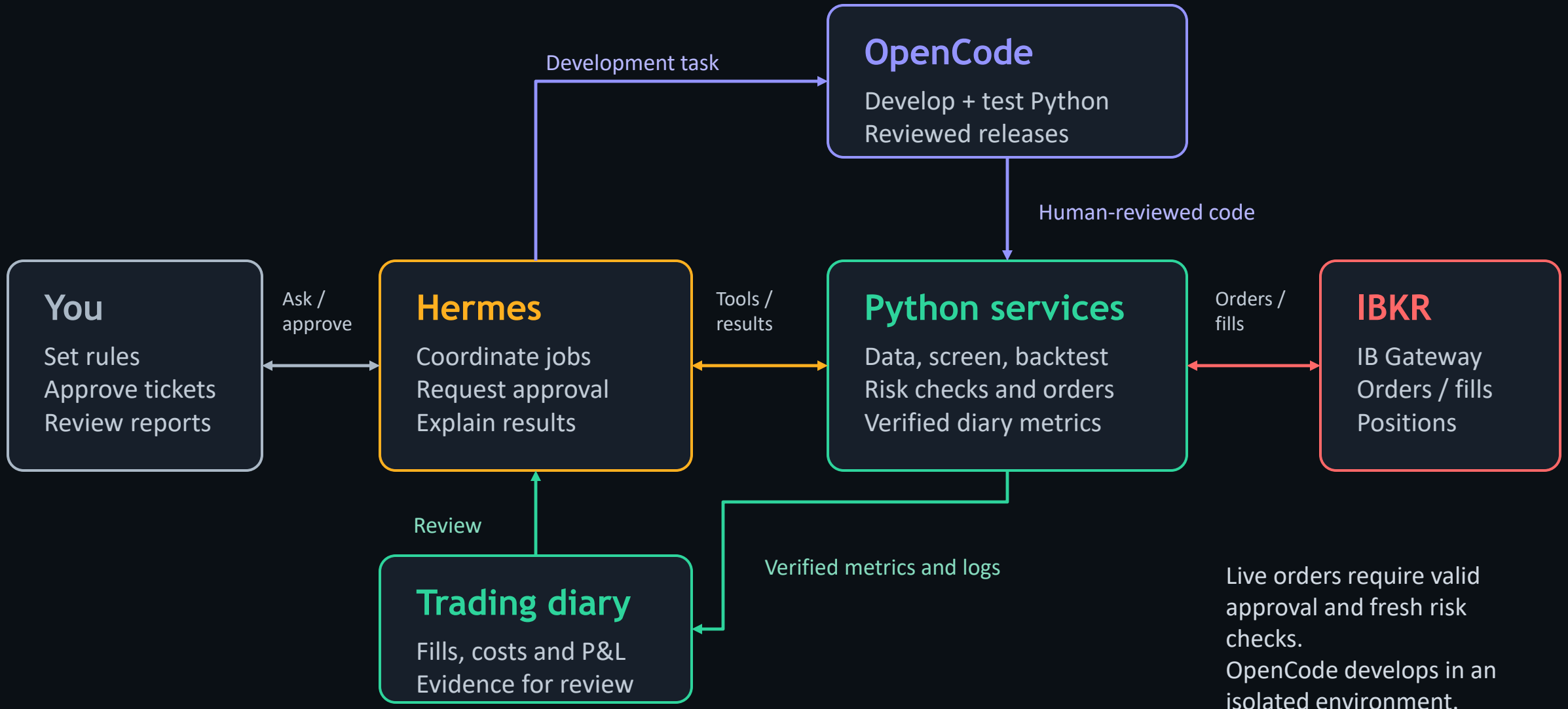
YOU

Decision authority

Approve strategy releases, risk limits and live tickets. Own the deployment decision and the independent kill switch.

Hermes and OpenCode trading workflow

Proposed integration with human approval and an isolated broker execution service.



Live orders require valid approval and fresh risk checks.
OpenCode develops in an isolated environment.

Where Cursor, OpenCode and Hermes sit in the trading workflow

CURSOR

Human review surface

Open the development branch in the editor. Inspect diffs, plots, notebooks and failing tests before any release.

OPENCODE

Hired implementer

Receives a scoped ticket. Writes screens, backtests and adapters in an isolated branch. No live broker socket.

HERMES

Workflow coordinator

Schedules approved jobs, explains diary facts, drafts the next OpenCode task and presents tickets. Never submits live orders.

YOU

Decision authority

Approve releases, risk limits and live tickets. Review in Cursor. Own paper-to-live promotion and the kill switch.

Hermes can write Python. That is not a reason to let it deploy.

CAPABILITY

Not the same as permission

Hermes can draft screens and adapters. A trading desk still separates who may edit research code from who may change live risk.

BLAST RADIUS

One process is one hop

If Hermes chats, edits files and deploys, a bad skill or scraped page can reach execution in one hop. OpenCode only touches a branch.

WRITE ≠ DEPLOY

Stop at the pull request

OpenCode writes code and tests. It does not copy a new submit_order.py onto the gateway because a chat message said deploy.

REVIEW

A diff, not a surprise

Cursor opens the branch. You see tests and plots before promotion. Hermes must not be the author and the release manager.

Hermes coordinates. OpenCode implements. You release. Python executes.

01 HERMES

Propose, do not implement

Files a scoped ticket from a diary finding or a new idea. Explains results. Must not edit live code or copy scripts onto the gateway.

02 OPENCODE

Write in a branch

Implements screens, backtests and adapters in an isolated worktree. Runs tests. Stops at a reviewable diff. No TWS socket.

03 YOU + CURSOR

Review, then promote

Open the branch in Cursor. Inspect diffs, plots and failures. Approve the release. Paper-to-live is a separate decision.

04 PYTHON SERVICE

Run versioned code only

Reviewed artifact computes signals and risk, checks approval and may talk to IB Gateway. No agent rewrites this process at runtime.

Project setup, skills and tool connections

One versioned project, with separate development, paper and live environments.

PROJECT

Repository structure

data/ and strategies/
backtests/ and execution/
risk/ and diary/
tests/ and reports/

OpenCode works in a development branch. Humans review releases before deployment.

INSTRUCTIONS

Reviewed skills

AGENTS.md records coding rules. Skills describe screening, backtest and diary procedures.

Review third-party skills and pin versions. Risk limits remain protected configuration.

CONNECTIONS

Restricted tools

Expose narrow tools for data, approved jobs and broker status through MCP or service APIs.

Keep approval verification and live submission in an isolated service. Use separate credentials.

Market data collection and validation

Required inputs

Historical OHLCV, fundamentals where needed, corporate actions and an exchange trading calendar.

Hermes coordination

Requests the approved data job and reports missing coverage or validation failures.

OpenCode development

Builds adapters and tests for missing bars, split events, duplicate rows and timestamp errors.

Data quality checks

Reject stale or incomplete inputs. Confirm symbols, time zones and consistent price adjustments.

Reproducible snapshots

Save source, retrieval time, as-of time and data version. Keep historical universe membership.

Failure policy

Block affected signals and tickets. Alert the user instead of silently filling gaps or inventing values.

Automated stock screening

Example rules below are illustrative. The human approves the universe and thresholds.

HERMES

Screening request

Run the approved US-stock screen after the daily data check.

Example: price above $5, adequate 20-day dollar volume, close above SMA200 and positive 3-month momentum. Rank eligible names.

OPENCODE

Rules and tests

Translate criteria into explicit Python filters. Test boundary cases and corporate-action handling.

Store thresholds in versioned strategy configuration. Calculate ranks consistently.

DELIVERABLE

Ranked watchlist

Hermes explains the Python results: ticker, rank, signal values, inclusion reasons and data timestamp.

Keep rejected names with reasons. A watchlist cannot authorize an order.

Strategy development and backtesting

Explicit strategy rules

Define signal timing, entry, exit, sizing and rebalance frequency before evaluating results.

OpenCode implementation

Build the Python backtest and fixtures. Check fills, costs, position accounting and invariants.

Hermes coordination

Runs approved experiments through the job runner and summarizes the saved results.

Realistic assumptions

Include commissions, spreads and slippage. State short-borrow costs and liquidity limits if relevant.

Performance measures

Return, maximum drawdown, Sharpe, Sortino, win rate, profit factor and benchmark comparison.

Bias and robustness

Use out-of-sample or walk-forward tests. Check look-ahead, survivorship and parameter sensitivity.

Strategy review and paper-trading validation

REVIEW

Evidence summary

Hermes compares saved strategy variants and the benchmark. Highlight drawdown, turnover, instability and missing evidence.

PAPER TRADING

Operational rehearsal

Run the reviewed release in a paper account. Exercise order handling, reconnects, cancellations and the trading diary.

ACCEPTANCE

Defined test gates

Require passing tests, reconciled positions and no unexplained duplicate orders. Agree evaluation criteria before the trial.

PROMOTION

Human release approval

You approve strategy and risk configuration before any small live trial. Paper fills do not prove real-market profitability.

Trade proposals, risk checks and approval

PROPOSAL

Structured ticket

Record account, instrument, side, quantity, order type, price limit, expiry and strategy version. Include a unique ticket ID.

PRE-TRADE RISK

Independent checks

Python checks funds, exposure, liquidity, current quotes and duplicate requests. Protected limits cannot change through chat.

HUMAN APPROVAL

Exact, expiring consent

Hermes presents the ticket and risk summary. A separate verifier checks your identity, ticket hash, expiry and one-time nonce.

SUBMISSION GATE

Fresh validation

Recheck risk and prices at submission. Changed, expired or already-used tickets fail closed and require a new approval.

IBKR order placement and position monitoring

EXECUTION SERVICE

Restricted broker access

Only the approved Python service connects to the authenticated TWS/IB Gateway session. It submits validated order tickets.

ORDER LIFECYCLE

Acknowledgements and fills

Track accepted, rejected, partial, filled and cancelled orders. Record broker IDs and events. Hermes delivers status alerts.

RECOVERY

Broker reconciliation

After a timeout or reconnect, query orders, fills and positions before retrying. Preserve ticket-to-order mapping across restarts.

SAFETY CONTROLS

Limits and kill switch

Enforce session rules, exposure caps and price collars. Halt new submissions on uncertainty. Keep manual broker access available.

Automated trading diary and performance review

Python supplies verified facts. Hermes provides a traceable narrative and review questions.

FACTS

Trading record

Record signal, decision time, ticket, approval, orders, fills, costs and exit reason.

Python computes realized and unrealized P&L. Preserve strategy version and broker identifiers.

REVIEW

Hermes commentary

Explain performance using the saved metrics. Separate observed facts from hypotheses.

Review execution quality, rule adherence, drawdowns and recurring errors. Cite source record IDs.

FOLLOW-UP

OpenCode tasks

Turn accepted findings into scoped development tickets.

Examples: test duplicate-order handling, improve cost attribution or check a signal discrepancy. Changes need tests and human review.

Controlled improvement and end-to-end demo

01 RESEARCH

Screen and backtest

Hermes requests the approved screen, explains the watchlist and runs a reproducible backtest through Python tools.

02 PAPER CYCLE

Approve, trade, review

You approve a paper ticket. Python validates and submits it, tracks fills and updates the diary. Hermes summarizes the outcome.

03 DEVELOPMENT

Investigate and test

Hermes drafts a task from a diary finding. OpenCode changes a development branch and runs regression tests and a fresh backtest.

04 RELEASE

Human-controlled change

You review the evidence and approve the release. Repeat paper validation. Live risk changes and orders require separate approval.